CS 407 Project Charter
AttireAI

Team:

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Problem Statement:

Online clothing shoppers frequently struggle with selecting the right size and style, leading to high return rates and customer dissatisfaction. While existing fashion apps offer basic size charts or generic style suggestions, they fail to provide a personalized, end-to-end solution that considers individual body measurements, skin tone, hair color, and occasion-specific needs. Our web application AttireAI can address this gap by combining AI-powered body measurement technology with intelligent style recommendations, allowing users to receive personalized outfit suggestions complete with pricing and direct purchase links. Unlike competitors such as Stitch Fix or Amazon StyleSnap, AttireAI offers live measurement capture, comprehensive color analysis, and a virtual try-on feature that generates realistic outfit previews on the user's own image.

Project Objectives:

- Develop a web application that allows users to input or capture their body measurements, including waist, chest, hip, inseam, and other relevant dimensions.
- Implement a live measurement feature using camera-based technology to automatically detect and record user body measurements.
- Integrate skin tone and hair color detection to enhance personalization of style recommendations.
- Create an occasion selection system where users can specify the context for their outfit (e.g., formal dinner, casual party, business meeting, date night).
- Develop a style preference module allowing users to select their preferred fashion styles (e.g., minimalist, streetwear, classic, bohemian).
- Build an AI-powered recommendation engine that suggests complete outfits based on user measurements, colors, occasion, and style preferences.
- Display outfit recommendations with pricing information and direct purchase links to partner retailers.

- Implement a user account system to save measurements, preferences, and past recommendations.
- Develop a VIP subscription tier that includes virtual try-on functionality, generating realistic preview images of recommended outfits for the user.
- Design a database to store user profiles, measurements, preferences, outfit history, and product information.

Stakeholders:

- **Users/Customers:** Online shoppers who want personalized clothing recommendations and accurate sizing guidance; fashion-conscious individuals seeking style advice for specific occasions.
- **Project Owners:** Yuanfei Song, Congtian Wu, Yichen Dai, Ekaterina Tszyao
- **Project Manager:** Ekaterina Tszyao
- **Software Developers:** Yuanfei Song, Congtian Wu, Yichen Dai, Ekaterina Tszyao
- **Development Manager:** The Project Coordinator (GTA) assigned to our team this semester.
- **Partner Retailers:** Clothing brands and online stores whose products will be recommended through the platform.

Deliverables:

- A responsive web application accessible via desktop and mobile browsers that provides personalized outfit recommendations based on user measurements, coloring, and preferences.
- **Frontend:** Built using React.js (or Next.js) for a modern, responsive user interface with TypeScript for type safety.
- **Backend:** Developed using Python with Flask/FastAPI to handle API requests and business logic.
- **Database:** MySQL to store user accounts, body measurements, style preferences, and recommendation history.
- **User Authentication:** Secure login system with support for email registration and social login options using Firebase.
- **External API Integration:** Connections to clothing retailer APIs, if possible, or affiliate networks for product data and purchase links.

CS 307 Project:

We were in the same CS 307 group.
Yuanfei Song, Congtian Wu, Yichen Dai, Ekaterina Tszyao:
GitHub link: https://github.com/nd1739655076/Judge_Everything

For CS 307, our team developed a web application called JudgeEverything, which was our first software development experience. The application serves as a user-driven review and rating platform that combines social media functionality with product comparison features. Users can create product profiles, set key parameters, rate and review products, and compare multiple items through consolidated scores and real-time rankings. The platform also supports rating non-product entries, such as events or memes, for entertainment purposes. We built the frontend using ReactJS and the backend using Firebase Cloud Functions, integrated with Firebase Database for data management. Additional features included a live chat system for user communication, a machine learning-based sensitive word checker, a personalized recommendation system based on user browsing history, and a separate manager platform for database administration and handling user requests.